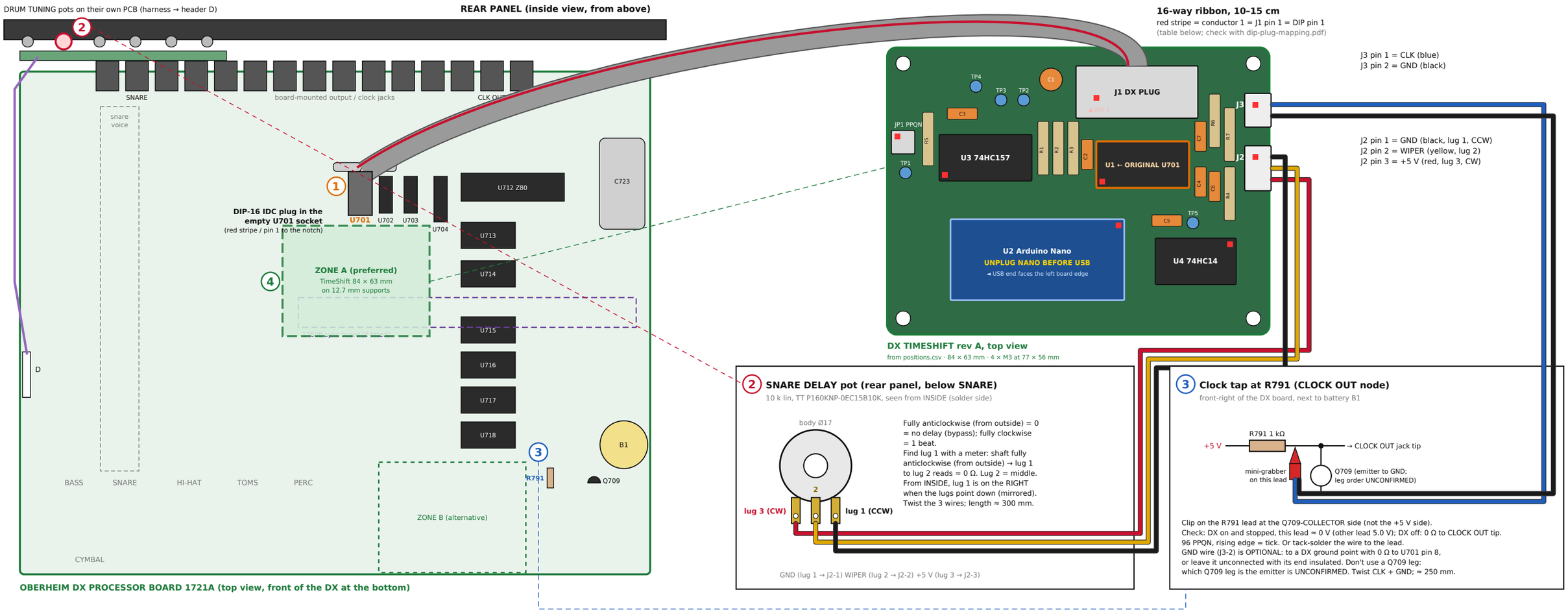
WIRING DIAGRAM — DX TimeShift in an Oberheim DX (board 1721A, U701)

Not to scale. DX board positions scaled from the Oberheim parts layout (±3 mm). Colours = suggested wire colours. Every pin number is also on the PCB silkscreen.



OBERHEIM DX PROCESSOR BOARD 1721A (top view, front of the DX at the bottom)

J1 ↔ RIBBON ↔ DIP-16 PLUG MAPPING (brief 3.5; source: hardware/kicad/tools/netlist.py)

J1 pin	cond.	DIP pin	U701 signal at that DIP pin	TimeShift path	J1 pin	cond.	DIP pin	U701 signal at that DIP pin	TimeShift path
1	1	1	CLR (tied +5V)	straight through to U1 pin 1	9	9	5	Q2 = FFA1 (bass)	straight through to U1 pin 5
2	2	16	VDD = +5V	+5 V supply (≤ 60 mA)	10	10	12	Q5 = FFA2 (SNARE)	INTERCEPTED via U3 mux (not U1)
3	3	2	Q1 = RST1 (bass)	straight through to U1 pin 2	11	11	6	D3 = bus D2	straight through to U1 pin 6
4	4	15	Q6 = RST2 (SNARE)	INTERCEPTED via U3 mux (not U1)	12	12	11	D4 = bus D5	straight through to U1 pin 11
5	5	3	D1 = bus D0	straight through to U1 pin 3	13	13	7	Q3 = FFB1 (bass)	straight through to U1 pin 7
6	6	14	D6 = bus D3	straight through to U1 pin 14	14	14	10	Q4 = FFB2 (SNARE)	INTERCEPTED via U3 mux (not U1)
7	7	4	D2 = bus D1	straight through to U1 pin 4	15	15	8	VSS = GND	GND
8	8	13	D5 = bus D4	straight through to U1 pin 13	16	16	9	CLK = GEN0*	straight through to U1 pin 9

Conductor n of the ribbon lands on J1 pin n; the DIP plug puts conductor n on the DIP pin shown. Orange = the three snare lines the TimeShift intercepts.

LEGEND

- +5 V (pot lug 3 / CW)
- GND (pot lug 1 / CCW, clock GND)
- pot wiper (lug 2)
- clock tap (R791 → J3-1)
- 16-way ribbon, red stripe = 1
- intercepted snare lines (10/12/15)
- suggested board zones

① U701 socket: lift the original MC14174B out (note the notch), fit it in U1 on the TimeShift, then push the DIP plug into the empty DX socket, pin 1 to the notch end. (Findings Q1: confirm U701 is socketed first.)
② New pot hole: rear-panel-drill-template.pdf. Do not touch the tuning-pot harness (header D).
③ Clock tap: mini-grabber on R791 (Q709-collector side). 10 kΩ + Schmitt input on the TimeShift side.
④ Board (84 × 63 mm) in ZONE A on 4 self-adhesive snap-lock supports (12.7 mm), J1 towards U701, ribbon ≤ 15 cm. Check lid clearance first (findings Q3: ≥ 32 mm incl. the Nano) and any 1760B sync board or Electrongate MIDI board in the area. ZONE B needs a 20–25 cm ribbon: use it only after the GEN0* scope check (findings Q4). Power comes through the ribbon (U701 pins 16/8): no extra wires; never use the +5M battery rail. Flash the Nano OUT of the board, or only with the DX switched on.
JP1 (PPQN): no shunt = 96 (plain DX); shunt on JP1 1-2 = 48 (D10); shunt on JP1 3-4 = 24 (D11).

PROJECT	SHEET 1 / 1 UNITS mm
DX TimeShift (snare) for Oberheim DX, board 1721A	
DRAWING	A3 landscape
Wiring diagram (pot, clock tap, ribbon)	
DWG NO. **DXTS-MECH-002**	SCALE **NTS**
REV **A**	DATE **2026-11-06**